Designing Data Collection Methods for Applying Learning Analytics in Resource Constrained Schools

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ABSTRACT
The use ofLearning Analytics (LA) has shown significant improvements in online learning environments. This paper intends to discuss the challenges of applyingLA in the context of elementary schools of India.Out of many challenges, we focus on solving the unavailability of school data in digital format. Most of the data in such schools is stored on physical mediums like notebooks. These schools also lack the critical resources like the availability of computing devices, internet and digitally literate man-power.As the primary outcome of this paper, we propose a data collection system, named as OSLA,which provides multiple ways of data collection in resource constrained environment. OSLA distributes the need of resources across different touchpoints to tolerate the unavailability of specific computing device, internet and digitally literate manpower.

Author Keywords
Learning Analytics; Elementary Schools; India; Resources; Resource Constrained; Learning Environment.

ACM Classification Keywords
• Human-centered computing~Collaborative and social computing systems and tools *• Social and professional topics~Socio-technical systems*

1. INTRODUCTION
The objective of this paper is to find opportunities and challenges in applying Learning analytics in resource constrained elementary schools of India. Here, by word 'resource', we mean the resources like availability of computing devices, internet and digitally literate man-power, which are required for applying learning analytics.The term "resource constrained" is used to include government as well as private schools who are facing shortage of such resources.

In the next section, we start by understanding the concept of Learning Analytics and why certain resources are necessary while using it. In the later part, we take closer look at the context of elementary schools and how learning analytics can help in improving its learning environment.

2. SECONDARY RESEARCH

2.1 Learning Analytics and Required Resources
Technology Enhanced Learning (TEL) is a well-established and deeply researched area [1]. Researchers, working under TEL, explore the use of ICT tools and techniques for improving learning environments. Learning Analytics (LA), a lately emerged sub-research area under TEL, specializes in analyzing educational data for the benefit of learner and tutor [1]. LA approaches the problem through user centric perspective while choosing methods from variety of fields depending upon user characteristics, context and its objectives. [2]

The research in Learning Analytics has shown promising results in the last decade [4]. One of most famous case study of Learning Analytics is the Course Signals application, developed in Purdue University [5]. Signals predicts students who may not perform well in the course by using factors like course performance, time on task and past performance. Signals was successful in increasing number of A and B grades awarded to students by 10%. Moreover, it was also successful in increasing retention of students in courses [4]. Another widely appreciated LA applications include SNAPP (Social Networking Adapting Pedagogical Practice) and GISMO. SNAPP analyses discussion threads in order to unearth hidden problems like learner isolation, biases in facilitator's engagement with learners and non-interacting groups [6].GISMO is an add-on library for Moodle whichvisualizes the logged data like student's visits to CMS, student's use of resources and submission of assignments in order to reveal meaningful insights to facilitator [7].

Learning analytics borrows its methods and techniques from statistics, information visualization, social network analysis and data mining, etc [2]. The method is chosen depending upon the nature of data, context and purpose of analysis. For example, data mining methods are used to

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predict unknown learner metrics or indicators. Commonly used data mining methods include classification, clustering and association rule prediction. As these analytical methods need higher computing abilities than the ability of a normal human, use of computers becomes necessary.

Most of the current LA tools as discussed above are built on top of already existing digital systems. Signals extends present university portal. SNAPP utilizes existing email clients and online discussion forums. GISMO is built on top of online learning management system, Moodle. It is evident that data required by these tools is readily available in the digital format. Additionally, all of these tools assume the availability of digital devices and internet connectivity. Just to give example, Signals uses course performance and past performance data. This data is captured using online tests which happen on university portals. Indirectly, Signals requires students and universities to have computers, mobile phones and internet to access system. Similarly, SNAPP and GISMO are also directly dependent on availability of computing and digital resources. Subsequently, these applications also assume the user of LA system to have the experience of using computers. In other words they expect user to be digitally literate. Therefore, availability of computing devices, internet connectivity and digitally literate manpower become critical resources to implement Learning Analytics system.

2.2 Condition of Elementary Schools of India

In India, 65% of children (age 6 -14) are enrolled in government schools [3]. More than 50% schools do not have enough teachers for teaching to all students [3]. Only 19.6% of schools have computers available for students [3]. Due to poor pupil-teacher ratio, teachers are not able to give personal attention to students and cannot keep track of students' progress efficiently. As a result, many students from these schools fail to exhibit even basic reading and arithmetic abilities because of poor learning environment.

Applying Learning Analytics in such schools can be beneficial for the improvement of students, teachers and learning environment itself. LA can be used to monitor and analyze student's progress. It can also help in identifying students at risk and allow teachers to intervene and take proactive measures. Insights provided by LA can allow teachers and students to scrutinize their own teaching or learning experiences so as to improve in the future. LA can also help in personalizing learning process of students depending upon their capacity, learning speed and current state.

In next section, we present our findings from the primary research. The aim of primary research was tounderstand data points and data collection practices and to understand the problems faced by teachers and students in such schools. It helped us in understanding ground level challenges and potential opportunities for LA.

3. PRIMARY RESEARCH

To recognize different data points in the school environment and to explore potential applications of learning analytics, three elementary schools from Maharashtra, India were visited. Out of three schools, 2 schools were run by municipality while other was run by a private body. All of these schools were facing paucity of one or more resources like teachers, reading rooms and computers. To tackle the shortage of resources, NGOs like Identity Foundation, Teach for India and Aakansha Foundation are assisting these schools.

The user study was done in two parts. First part comprised of conducting semi-structured interviews in school environment using contextual inquiry technique. In total, 4 elementary school teachers, 1 NGO senior officer and 3 NGO volunteers were interviewed. The second part of user studies involved shadowing sessions in a 4th grade classroom for continuous 4 days to manually collect sample data related to pre-recognized data points and to further explore new data-points which were missed during interviews.

Figure 1 showcases categories of data points that were recognized during user studies. This gives idea about what kind of data exists about different stakeholders of school environment. From these data points, it was observed that most of the data, which is currently being recorded, is ordinal in nature. Very few data points, like teacher's feedback about student's progress,were nominal in nature.

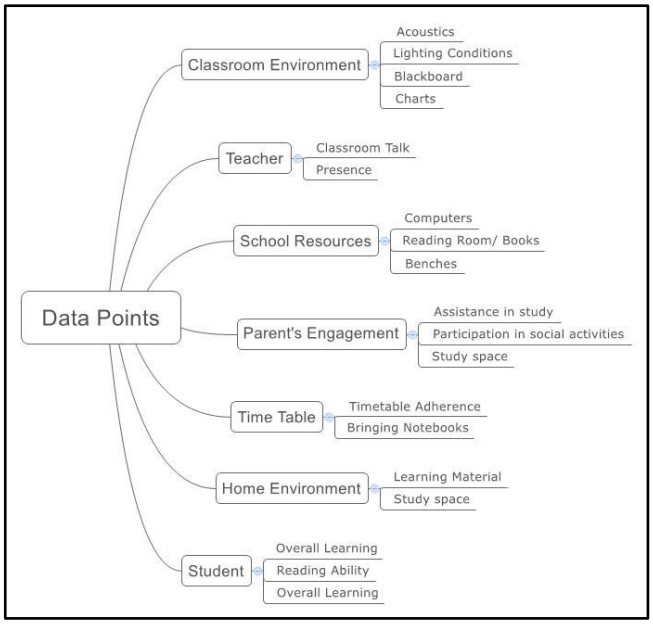


Fig 1: Categories of Data Points found in primary research

The same data points are collected repeatedly over the academic semester. For example, attendance data is collected on daily basis, while data like number of enrolled students is recorded on monthly basis. Similarly, reading ability test score and subject wise marks are recorded on

every quarter and every semester respectively. This shows that the need of data collection in schools is recurring.

The data collected by teachers at school level is recorded in the registers and notebooks. In case of NGOs, volunteers are also asked to manually enter and store the data in computers in tabular format. These volunteers are trained separately by NGOs to use computer and MS office. The collected data is analyzed by a senior officer.

During interviews, teachers also complained that they are not able to utilize their available classroom time due to training, handling multiple classrooms simultaneously and preparation of school events and holidays.

Findings from secondary and primary research indicate that most of the data in school environment is recorded on physical medium and availability of resources required for Learning Analytics cannot be guaranteed. Hence, there is need to design ways of data collection which can work in offline as well as online environment. Additionally, one should also try to minimize the efforts of manual data entry.

Before proposing new ways of data collection, in next section we critically review existing data collection tools for identifying common workflows, design patterns and building a state-of-the-art feature repository.

4. STUDY OF EXISTING WAYS OF DATA COLLECTION

The data collection tools can be categorized in offline and online tools. For understanding online data collection tools, SurveyMonkey and Google Forms were reviewed. For understanding offline data-collection tools, U-DISE initiative and OMR softwares were reviewed.

4.1 Review of Online Data Collection Tools

One of the prime feature of Online data collection tools like SurveyMonkey[8] and Google Forms[9] is that they allow their users to create their own questionnaires and collect data for almost any purpose. Additionally, learning time of questionnaire creation functionality is low due to its well-designed interface in terms usability and micro-interactions. For example, while adding new questions, most used question type is pre-selected along with actionable hint text, which decreases user's effort. Whenever user hovers mouse cursor on choice or question, its affordance changes to represent editing possibility. Micro-interactions like this make learning very easy even for a beginner user. Along with well-designed questionnaire creation functionality, SurveyMonkey allows user to customize already created questionnaire templates to create new ones, thereby saving user's time and efforts. Apart from these upsides, these tools do not allow to collect data without internet connectivity. Additionally, it is not possible to conduct longitudinal surveys using online data collection tools as identity of participant is not maintained across multiple surveys. User is required tomanually join the data from multiple surveys together for collective analysis.

4.2 Review of Offline Data Collection Tools

U-DISE (Unified District Information System for Education) is an Indian government initiative for nation-wide collection of school data and its analysis [10]. U-DISE distributes a software and a paper-based form of fixed format to all schools. School headmasters are asked to fill this form and submit to regional offices. Due to paper-based data collection form, School teachers are not required to be digitally literate to collect and submit the data. Data entry operators fill this data onto software as well as ensure correctness of data. After collecting data, the data is analyzed to create report for each cluster, block, district and state. The major drawback of U-Dise's process of data collection is that data collection form (DCF) has fixed parameters. Alteration to DCF also requires upgradation of software and its database schema.

While U-DISE depends manual data entry operators, OMR (Optical Mark Recognition) technique can be used to capture hand-marked data into digital format without manual data entry operators. Using available OMR softwares, one can design an OMR compliant questionnaire. This questionnaire can be printed using easily available black and white printers. After collecting data, filled responses are needed to be scanned using scanner and same OMR software can be used to process scanned images and fetch data. Due to its cost effectiveness and accuracy in collecting mass data, OMR technique is vastly used in conducting closed question examinations like entrance exams. Collecting textual/ descriptive data using OMR is difficult and tedious. Additionally, OMR can be implemented using readily available digital devices like a personal computer and a scanner.

Along with decreasing efforts of manual data entry, OMR also decreases digital literacy barrier. The capability of OMR technique to capture data from human readable printed forms encapsulates the need of data collector to have knowledge about computer, OMR software or the technique itself. Additionally, availability of computer or scanner is required only while creating OMR sheet and scanning filled responses. But it is not required during data collection.

To understand common workflows and problems related to OMR technology, we reviewed contemporary OMR applications including FormReturn [12], Socrato [11], FormScanner [13] and SDAPS [14]. One of the important feature exhibited by Socrato software is that it gives facility to generate bubble sheets. For giving SAT test offline, students can download bubble sheets, give test and upload scanned bubble sheet online to get score. This approach of auto-generating OMR compliant questionnairescan be useful in resource constrained school environment, as it removes the efforts of designing one. Additionally, this approach creates a distributed system where user is not required to have OMR software installed on

his/hercomputer in order to collect data. Subsequently, user can use OMR technology without being trained for OMR software.

Besides this upside, even though OMR techniques is under practice for more than two decades, OMR softwares have failed to evolve in terms of usability. For example, users need higher learning time than online data collection for the functionality of creating questionnaires. OMR softwares expect their users to decide placement of barcode, every question and its choices. Some OMR softwares lack a graphical user interface. They expect users to use a command line user interface for fetching data from scanned images. Moreover, there does not exist any common standard for creating OMR documents. Different OMR softwares use different shape (circular and rounded rectangle) to represent options. As a result, OMR questionnaires created using different softwares are not compatible with each other. Hence, user have to stick to a single OMR software throughout the whole process of data collection.

Availability of scanner is also a major hurdle in implementing OMR technology in resource constrained schools. However, nowadays the computing power of smartphones have increased. Smartphones are equipped with more powerful cameras. It is now possible to run heavy duty image-processing algorithms on smartphones. There are many applications like Google Drive and CamScanner who offer scanning of pages using phone camera. Hence, if implemented, it might be possible to use smartphones for scanning OMR sheets using their camera.

One cannot expect teachers to spend lot of time in learning the software, as they are already loaded with lot of academic work. Additionally, some teachers might not be digitally literate enough to understand and carry out complex workflows.

We conducted adata collection activity was conducted in Krantivir Vasudev Balwant Phadke Vidyalaya with 125 student participants. The aim of the study was to understand the complete workflow of OMR data collection process and challenges while using OMR with teachers and school environment. The data collection activity resulted in discovery of two more workflows which must be considered while implementing OMR technology. First workflow is of testing OMR questionnaire before usage. It was found that it is necessary to test designed OMR before actually collecting data from real participants. This involves checking whether OMR software is able to fetch data from a scanned image of sample response. The second workflow is of error correction. Many times OMR sheets are not scanned properly which leads to errors like undetectable barcode, section bar code and answer. In this case, user needs to manually enter respondent's answer by looking at the section scanned image.

To summarize this section, OMR technique can be used resource constrained schools but its workflows and user interface need to be redesigned for simplicity and usability.In the next section, considering needs and design ideas discussed in previous sections, we propose a survey development and learning analytics platform.

5. OSLA: PROPOSED SOLUTION

As this paper focused towards designing of data collection methods, the needs related to analysis of data and potential areas for applying learning analytics are not discussed here. Considering needs related to data collection methods and analysis of data, we designed the concept of OSLA(Open School Learning Analytics). As its name suggests it is a common platform for schools teachers, researchers and NGOs to openly share data and its analysis with each other. OSLA helps in capturing micro-level data of students, teachers and school environment. Unlike other platforms, OSLA relates data of students and their schools collected from multiple surveys thereby allowing its stakeholders to draw collective insights from the analysis. .

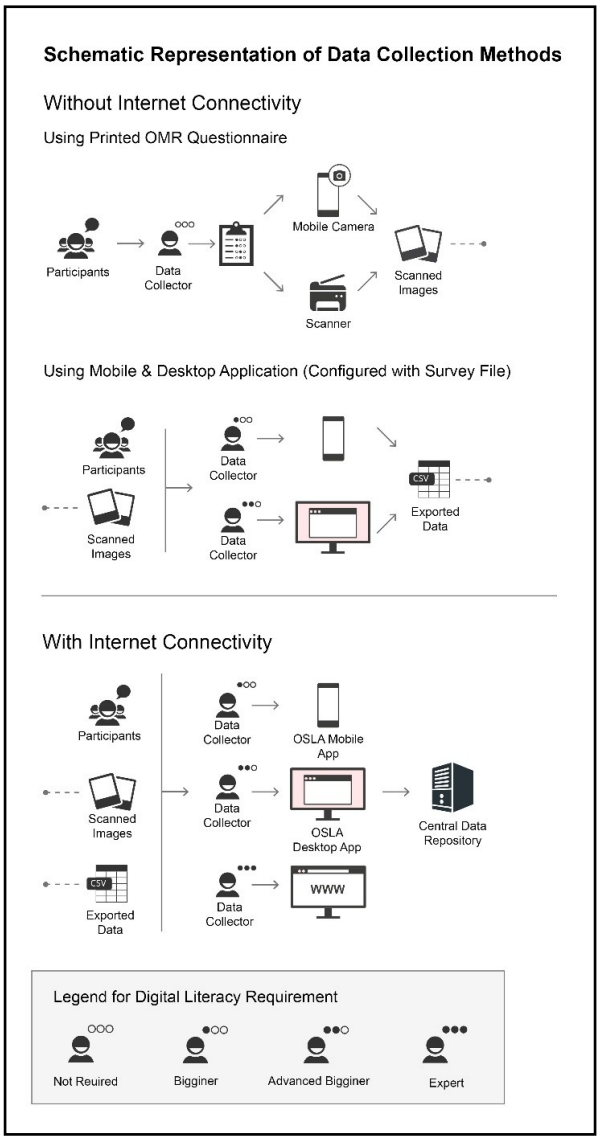


Fig 2: Data Collection Methods offered by OSLA platform

OSLA is designed to work as a distributed system for data collection. OSLA offers a mobile application, desktop application and a web portal as its touch points. By word 'distributed', we meant that OSLA tries to minimize the dependency on resources at a single touch-point at a single point of contact. This approach allows OSLA to build tolerance towards unavailability of any particular resource at given amount of time. Multiple data collection methods offered by OSLA are shown in Figure 2.

5.1 Touch-Points

5.1.1 OSLA Mobile Application

The mobile application will primarily be used for data collection. It is expected to be installed on data collector's personal smartphone if they already have one. In case of availability of internet connectivity, data collector can directly be fill data directly on mobile application. If internet connectivity is not available, data collector can configure survey into mobile application using a survey file, which can be downloaded from OSLA web portal. After configuration, mobile application can be used to collect data without internet connectivity. The collected data is stored on local memory and can be exported to suitable format like CSV whenever required. This data will get synced to server whenever phone gets connected to internet. Otherwise, exported csv file can be uploaded to central server using other online OSLA mobile application or desktop application or web portal.

In addition to directly collecting data, mobile application can be used to fetch data from printed OMR sheets. If configured with survey file, even without internet connectivity, OSLA mobile application can be used to take photograph of OMR sheet using phone camera and process the photograph to fetch data. Additionally, mobile application can also be used for correcting error in the processed data.

5.1.2 OSLA Desktop Application

OSLA Desktop application and mobile application have same purpose that is to allow easy collection of data with or without availability of internet connectivity. The OSLA Desktop application as a touch-point is necessary because it was observed in user studies that most of the municipality schools have at least one computer which is provided by municipality in order to carry out administrative tasks. Additionally, one cannot guarantee that teachers will have smartphone for using mobile application for data collection. The functional features of desktop and mobile application are same.

5.1.3 OSLA Web Portal

The primary function of OSLA Web Portal is to allow creation OMR sheet and analysis of data. OSLA web portal can also be used for administrative tasks like creating data collector accounts, assigning data collection responsibilities and keeping track of collected data. Web portal can be accessed using web-browser without needing to install OSLA mobile and desktop application. Web portal can also be used for uploading collected data directly on the central server. It is required while uploading photographs of printed bubble sheets or exported CSV data.

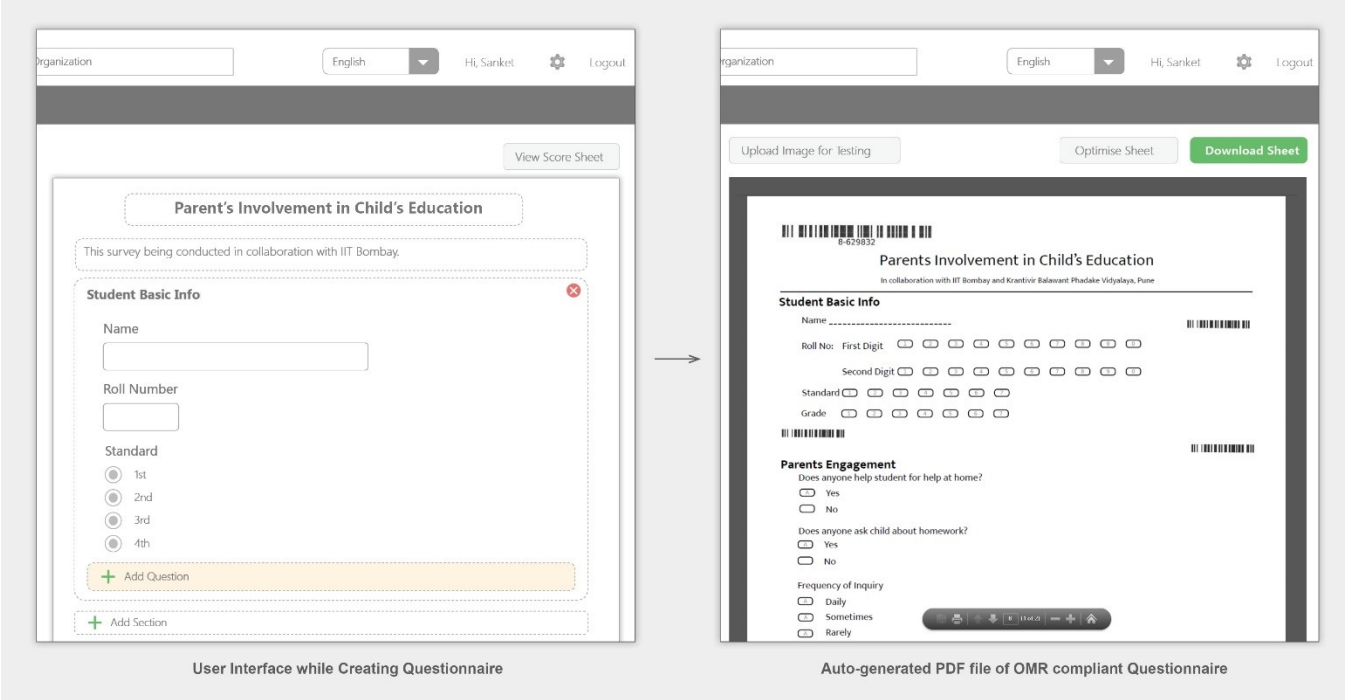


Figure 3: Left side shows parts of UI while creating questionnaire. Right side shows part of UI where OMR compliant questionnaire is autogenerated

5.1.4 Printed OMR Questionnaires

Printed OMR sheets will be used by data collectors who do have personal smartphone or who are not comfortable with using smartphone. After filling responses, these printed sheets can be processed using OSLA mobile application installed on the smartphone of other data-collectors. Otherwise, data collectors can submit these sheets to survey administrator. Survey Administrator can scan these sheets using scanner and upload to web-portal for fetching the data.

5.2 Key Features

OSLA combines design patterns of online survey development applications with the workflows of OMR softwares. The result is a simplified workflows of OMR questionnaire creation, conducting survey and error correction. In this way, OSLA is expected to have less learning curve than traditional OMR softwares.

5.2.1 Designing OMR Questionnaires

For using OMR sheets, OSLA provides facility of auto-generating OMR sheet without requiring user to worry about placement of barcode and content. Using created questionnaire, OSLA auto generates multiple templates of OMR sheet [Figure 3]. User can select the template as per requirement. Additionally, if required, user can do sight modifications in arrangement of questions and orientations of choices.

For creating questionnaires faster, users can also use existing questionnaires to build on top of it. Users can also utilize section templates like student basic information and teacher information provided by OSLA platform. As OSLA is designed to collect any type of ordinal data, one can create survey to capture any type of existing ordinal data like attendance of students, exam grades and subject-wise marks. OSLA enforces to use identifier for school, teachers and student while creating survey so that collected data can be related back after uploading.

5.2.2 Collecting Data with or without Internet

Using OSLA platform, survey can be conducted on multiple channels, with or without internet connectivity. OSLA allows survey to be conducted using printed OMR sheets, its mobile application, desktop Application and web-portal.

After creating questionnaire, auto generated OMR sheet can downloaded as PDF file and printed using black and white printer. Later on data collector can collect the data using these sheets and submit filled responses back to survey administrator. OSLA provides multiple ways of fetching data from printed OMR sheets.

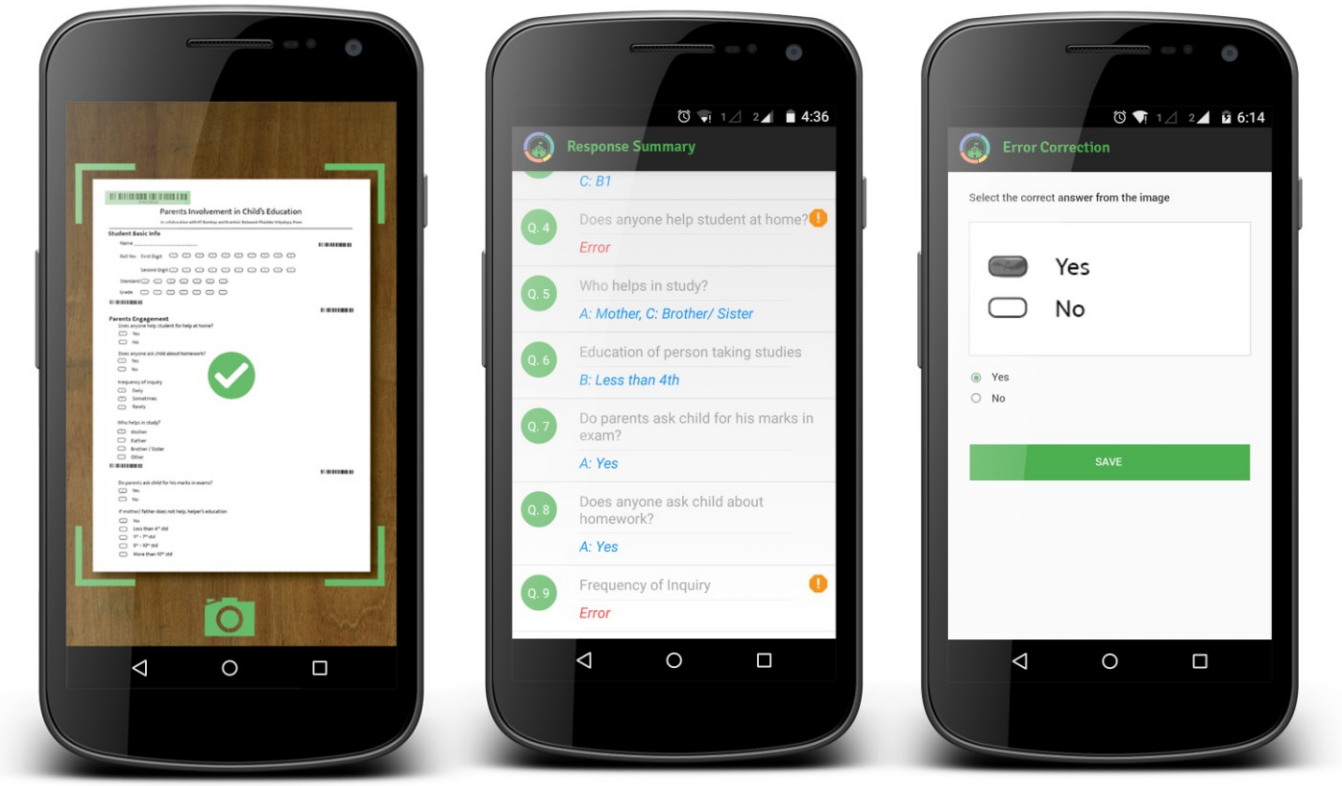


Figure 4: From left to right, 1st Screenshot shows interfacefor scanning OMR sheet using OSLA mobile application. 2nd Screenshot shows fetched data after processing scanned image along with errors.3rd Screenshot shows interface for correcting errors.

OMR sheets can be scanned using regular scanner. The resulting images can then be processed using OSLA's mobile app, desktop app or website. OSLA's smartphone application can be used to take photograph of printed OMR sheet and process the response. [Figure 4]

OSLA creates a ".srv" configuration file for a survey. Data collectors can download this survey file from web-portal. This file can be configured in OSLA mobile or desktop application. This enables application to be able to conduct survey without internet connectivity. Additionally, it also allows OSLA application to process OMR sheet responses.

Whenever, application gets connected to internet, it automatically uploads data to central server. Otherwise, data can be exported in CSV format and can be submitted to central server through any other OSLA application having internet connectivity or website.

6. USER FEEDBACK

The purpose of user evaluation for this project was to understand conceptual, usability and feasibility related issues faced by user while new functionalities provided by OSLA platform. Clickable prototypes of mobile application and web-portal were developed. The mobile prototype was developed on android platform. For web-portal, interactive PDF was created using screenshots.

Evaluation was be conducted in two parts. First part involved heuristic evaluation of prototype using with 4 different Interaction Designers. It helped in getting expert's feedback on usability of product. Each session lasted around 1 hour and 30 minutes. Second part involves testing of prototype in actual context with 4 users.

6.1 Protocol for User Testing

Before starting evaluation, participants were given a small training session. In this training session, the purpose of OSLA platform and its key features were explained. After training, participant were asked to perform task as per his/ her role. For example, if a participant was recruited for data collector role, he/ she was be asked to perform prime tasks of data collection role. In the task, participants was given a specific task to perform. Participants were provided a smartphone or laptop depending upon type of task. During the evaluation test, participants were observed to collect data about evaluation criteria. For OMR workflow related tasks, user were asked to perform task after giving demo of same task on FormReturn OMR software. After performing task, participants were interviewed about their feedback and overall experience.

6.2 Findings

Users found proposed data collection process very easy. For creating OMR complaint questionnaire, users found OSLA very easy as compared to the demonstrated OMR software due to less efforts and relatively simple user interface. Users liked the fact that data for students can be co-related. Hence, it is possible to track longitudinal growth of students. Users suggested that even health and sport related data can be added. This can be considered as positive feedback as users were suggesting new ways of using OSLA platform.

All users were able to capture OMR Responses by taking Photographs. Some users thought that using OMR sheets for capturing responses is mandatory which was not intended. The reason behind this could be the overall emphasis on OMR related tasks in the evaluation.

Users needed guidance while correcting errors for first time. All of them were successful in second attempt. To reduce the entry barrier, user should be provided with help about how-tos of using OSLA. It can be provided using embedded tutorials, tooltips, FAQs.

One of the important concern was that how OSLA will authenticate identity of survey participants. Data collector can submit fake data without even taking data from real participants. It was also suggested that Parents/ Students should get option to keep track of their data. There lies an ethical concern that learner's data will be used for unknown purpose without his/her consent. Users seemed OK with the use of the English language. However, user's responses were based on short interaction with concept. Users might face language problems while having long Interaction.

7. CONCLUSION

In this project, for availing learning analytics in resource constrained schools of India, we propose a system which is designed to tolerate unavailability of resources and to provide multiple ways of accessing, collecting and analyzing data. The results of evaluation show user's positive feedback towards UI flow of data collection methods in resource constrained schools. However, users also mentioned privacy and ethical concerns related storage and sharing of data. Additionally, implementing OMR library for mobile devices is still a challenge which needs to be solved. Even though users did not complain about using English language for interface, there is need to research issues related to its use in detail. Moreover, currently only mobile application part of proposed eco-system is implemented.

8. REFERENCES

[1] Ferguson, Rebecca (2012). Learning analytics: drivers, developments and challenges. International Journal of Technology Enhanced Learning, 4(5/6) pp. 304–317.

[2] M.A. Chatti, A.L. Dyckhoff, U. Schroeder, and H. Thüs , A Reference Model for Learning Analytics , International Journal of Technology Enhanced Learning (IJTEL).

[3] ASER 2014 - Annual Status of Education Report, January 2015

[4] Mark van Harmelen and David Workman, JISC CETIS Analytics Series: Vol.1 No. 3. Analytics for Learning and Teaching, 2012

[5] Steve Tally, Signals tells students how they're doing even before the test, University News Service, Purdue University, 2009. Retrieved from: http://www.purdue.edu/newsroom/students/2009/090827ArnoldSignals.html

[6] S. Dawson, A. Bakharia, E. Heathcote, ''SNAPP: Realising the affordances of real-time SNA within networked learning environments'' (2010)

[7] Riccardo Mazza and Christian Milani , GISMO [Software], Retrieved from http://gismo.sourceforge.net

[8] SurveyMonkey [Webapp], Retrieved from https://www.surveymonkey.com/

[9] Google Forms [Webapp], Retrived from https://www.google.co.in/forms/about/Google

[10] U-DISE [Software], Retrieved from http://www.dise.in/udise.html

[11] Socrato [Webapp], Retrieved from http://www.socrato.com

[12] FormReturn [Software], Retrieved from https://www.formreturn.com/

[13] FormScanner [Software], Retrieved from http://sourceforge.net/projects/formscanner/

[14] SDAPS [Software], Retrieved from http://sdaps.org/Documentation/GUI